

Software Testing – Question Bank

Practical 1 – SDLC and STLC

1. Write the phases of SDLC.
2. Write the phases of STLC.
3. Write the testing activities performed during the Requirement phase.
4. Write the testing activities performed during the Design phase.
5. Write the testing activities performed during the Testing phase.
6. Differentiate between SDLC and STLC.
7. Write the relationship between SDLC and STLC.
8. For an Online Shopping System, identify the testing activities performed during different SDLC phases.
9. Write any four activities performed by a tester during STLC.
10. Write a simple SDLC–STLC flow for a Login System.

Practical 2 – Test Plan, RTM, Scope and Testing Strategy

1. Write the main contents of a Test Plan.
2. Prepare a simple Test Plan for a Login Module.
3. What is a Requirement Traceability Matrix?
4. Prepare a simple RTM for a Login Module having 3 requirements.
5. Write the purpose of an RTM.
6. Write the meaning of Testing Scope.
7. Prepare the testing scope for an Online Shopping Application.
8. Write **In-Scope** and **Out-of-Scope** items for a Login Module.
9. What is a Testing Strategy?
10. Write a simple testing strategy for a Registration Module.
11. Differentiate between Test Plan and Testing Strategy.
12. Write the relationship between Requirement, Test Case and RTM.

Practical 3 – Requirement Review, Walkthrough, Inspection and Peer Review

1. What is Requirement Review?
2. Write the purpose of Requirement Review.
3. Write 5 activities performed during Requirement Review.
4. What is a Code Walkthrough?
5. What is Inspection?
6. What is Peer Review?
7. Differentiate between Walkthrough and Inspection.

8. Differentiate between Review and Peer Review.
9. Identify possible defects in a sample Login requirement.
10. Write 5 questions that a tester should ask during requirement review.
11. Write the steps involved in reviewing a requirement document.
12. Give an example of a requirement that is incomplete or ambiguous.

Practical 4 – Functional, Non-Functional, Positive and Negative Test Cases

Login

1. Write 5 functional test cases for a Login Module.
2. Write 2 positive test cases for Login.
3. Write 3 negative test cases for Login.
4. Write test cases for blank username and password.
5. Write test cases for an incorrect password.
6. Write a test case to verify the Login button.

Registration

7. Write 5 functional test cases for a Registration Form.
8. Write positive and negative test cases for an Email field.
9. Write positive and negative test cases for a Mobile Number field.
10. Write a test case for mandatory fields.

Payment

11. Write 5 test cases for an Online Payment Module.
12. Write a positive test case for successful payment.
13. Write a negative test case for failed payment.
14. Write a test case for an invalid card number.
15. Write a test case for insufficient balance.

Non-Functional

16. Write 3 non-functional test cases for a Login System.
17. Write test cases for the performance of a Login System.
18. Write test cases for usability of a Registration Form.
19. Differentiate between functional and non-functional testing.
20. Differentiate between positive and negative testing.

Practical 5 – Defect Identification and Bug Report

1. What is a defect?
2. Identify the defect from a given failed test case.

3. Write a professional bug report for an incorrect Login password message.
4. Write a bug report for a Submit button that does not work.
5. Write a bug report for an incorrect payment amount.
6. Write a bug report for an application crash.
7. Write the fields included in a bug report.
8. Write the difference between Severity and Priority.
9. Give an example of High Severity and High Priority.
10. Give an example of High Severity and Low Priority.
11. Write reproduction steps for a Login defect.
12. Write Expected Result and Actual Result for a failed Login test.
13. Write a complete bug report using:
 - Bug ID
 - Title
 - Steps to Reproduce
 - Expected Result
 - Actual Result
 - Severity
 - Priority

Practical 6 – Manual Test Cases and Test Execution Report

1. Write 5 manual test cases for a Login Module.
2. Write 5 manual test cases for a Registration Module.
3. Write test cases using different input conditions.
4. Write valid and invalid test cases for a Mobile Number.
5. Write valid and invalid test cases for an Email field.
6. Write a test case execution table for 5 test cases.
7. Prepare a test execution report for the given test cases.
8. Mark the following test cases as Pass or Fail based on the given actual result.
9. Write the fields of a Test Execution Report.
10. What is a Test Summary Report?
11. Prepare a simple Test Summary Report.
12. Calculate the number of Passed and Failed test cases from a given execution result.
13. Write the difference between Expected Result and Actual Result.

14. Write a test case with Test Case ID, Test Data, Expected Result and Actual Result.

Practical 7 – BVA and Equivalence Partitioning

Boundary Value Analysis

1. A field accepts **age from 18 to 60**. Write boundary test values.
2. A marks field accepts **0 to 100**. Write boundary values.
3. A password accepts **8 to 15 characters**. Write boundary test cases.
4. A quantity field accepts **1 to 50**. Write boundary values.
5. A mobile number requires exactly **10 digits**. Write suitable test values.
6. Explain Boundary Value Analysis with an example.

Equivalence Partitioning

7. Divide age **18–60** into valid and invalid equivalence classes.
8. Divide marks **0–100** into valid and invalid equivalence classes.
9. Divide password length **8–15** into valid and invalid equivalence classes.
10. Divide quantity **1–50** into equivalence classes.
11. Explain Equivalence Partitioning with an example.
12. Differentiate between BVA and Equivalence Partitioning.
13. Write test cases using BVA for a Registration Form.
14. Write test cases using Equivalence Partitioning for a Login Form.

Practical 8 – Bugzilla and TestLink

Since there is **no PC**, ask students to write the **workflow/process**, not actually operate the software.

1. What is Bugzilla?
2. What is TestLink?
3. Write the steps for reporting a defect in Bugzilla.
4. Write the important fields required while creating a defect in Bugzilla.
5. Write the steps for assigning a defect.
6. Write the steps for updating a defect status.
7. Write the Bugzilla defect life cycle.
8. What is Test Case Management?
9. Write the steps for creating a test case in TestLink.
10. Write the steps for executing a test case in TestLink.
11. Write the meaning of Pass, Fail and Blocked in test execution.

12. Write the workflow:

Requirement → Test Case → Test Execution → Result → Defect

13. Differentiate between Bugzilla and TestLink.

14. Write how a failed test case is connected with a defect.

Practical 9 – Black-Box, White-Box, Integration and System Testing

1. What is Black-Box Testing?
2. What is White-Box Testing?
3. Differentiate between Black-Box and White-Box Testing.
4. Give an example of Black-Box Testing for Login.
5. Give an example of White-Box Testing for Login.
6. Write 5 Black-Box test cases for an Online Shopping System.
7. What is Integration Testing?
8. Write integration test cases for Cart and Payment.
9. Write integration test cases for Login and User Profile.
10. What is System Testing?
11. Write 5 system test cases for an Online Shopping Application.
12. Differentiate between Integration Testing and System Testing.
13. Identify the suitable testing type for a given situation.
14. Write a comparison of Black-Box, White-Box, Integration and System Testing.

Practical 10 – Automation Testing and Regression Testing

1. What is Automation Testing?
2. What is Regression Testing?
3. Differentiate between Manual Testing and Automation Testing.
4. Identify which functionalities of an Online Shopping Application are suitable for automation.
5. Identify which Login test cases should be automated.
6. Write 5 test cases suitable for automation.
7. Write 5 test cases that can be included in Regression Testing.
8. Why is Login suitable for automation testing?
9. Why is Regression Testing important after software changes?
10. Write an example of a regression testing scenario.
11. Identify suitable automation candidates from:

- Login
- Search
- Payment
- Static text
- Frequently repeated functionality

12. Write the basic process of Automation Testing.

13. Write the difference between Automation Testing and Regression Testing.

Practical 11 – Python Automation: Navigation, Forms, Workflows and Links

1. Write the basic Selenium-Python steps for opening a website.
2. Write a Selenium-Python statement to open a URL.
3. Write a Selenium-Python statement to navigate to another page.
4. Write the steps for testing a multi-page workflow.
5. Write the automation steps for:

Home → Login → Dashboard → Logout

6. Write the automation steps for:

Home → Search → Product → Cart → Payment

7. Write test cases for website navigation.
8. Write test cases for form submission.
9. Write test cases for checking links.
10. What is a broken link?
11. Write test cases for links on a website.
12. Write a regression testing scenario using Selenium.
13. Write the difference between manual and automated execution.
14. Write the sequence of actions for automating a Registration Form.

Practical 12 – AI-Powered Software Testing and Generative AI

1. What is AI-powered software testing?
2. Write 5 applications of AI in software testing.
3. How can Generative AI generate test cases?
4. Write a prompt that can be used to generate Login test cases.
5. Write a prompt to generate test data for a Registration Form.
6. How can Generative AI help in defect prediction?
7. How can AI help in requirement analysis?
8. How can Generative AI generate Selenium test scripts?

9. Write 5 test cases that can be generated using AI for a Login Page.
10. Write sample test data that AI could generate for a Registration Form.
11. Write the advantages of AI-powered testing.
12. Write limitations/challenges of using Generative AI for testing.
13. Explain how AI can support Software Quality Assurance.
14. Write the flow:

Requirement → AI Analysis → Test Case Generation → Test Data → Automation Script → Testing

Practical 13 – Python, Selenium, IDE and Browser Driver Configuration

1. What is Python?
2. What is Selenium WebDriver?
3. What is VS Code?
4. What is PyCharm?
5. What is a browser driver?
6. Why is a browser driver required in Selenium?
7. Write the basic steps for setting up Selenium with Python.
8. Write the steps for installing Selenium.
9. Write the purpose of the Selenium library.
10. Write the basic structure of a Selenium-Python program.
11. Write the difference between Python and Selenium.
12. Write the role of IDE in automation testing.
13. Write the role of WebDriver in Selenium.
14. List the components required for Selenium testing using Python.

Practical 14 – Selenium Browser Navigation, Forms and Web Elements

1. Write Selenium code to open a website.
2. Write Selenium code to enter text into a textbox.
3. Write Selenium code to click a button.
4. Write Selenium code to clear a textbox.
5. Write Selenium code to locate a web element.
6. What is a Web Element?
7. What is a Locator?
8. Write different Selenium locator types.
9. Write the automation steps for a Login Form.

10. Write the automation steps for a Registration Form.
11. Write a test case to verify a Login button.
12. Write a test case to verify a textbox.
13. Write a test case to verify form validation.
14. Write the sequence:

Open Browser → Open URL → Locate Element → Enter Data → Click → Validate

15. Write Selenium-Python code for a simple form interaction.

Practical 15 – Automated Login, Authentication, Assertions and Exception Handling

1. Write the steps for automating a Login functionality.
2. Write a Selenium-Python program structure for Login.
3. Write a positive Login test case.
4. Write a negative Login test case.
5. Write test cases for invalid username.
6. Write test cases for invalid password.
7. Write test cases for blank username and password.
8. What is authentication validation?
9. What is an assertion?
10. Why are assertions used in Selenium testing?
11. Write an assertion to verify a page title.
12. Write an assertion to verify successful Login.
13. What happens when an assertion fails?
14. What is exception handling?
15. Why is exception handling required in Selenium?
16. Write the basic structure of **try-except** for Selenium.
17. Write a test scenario where an element is not found.
18. Write a Login automation flow with validation:

Open Website → Enter Username → Enter Password → Click Login → Verify Result

19. Write one **PASS** and one **FAIL** Login test case.
20. Write a Selenium Login test case containing **authentication validation, assertion and exception handling**.

